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Composable infrastructure enabled by heterogeneous architecture, delivered by CXL based cached switch SOC and extensible via cxloverethernet (COE) protocols

Abstract

Described herein are systems, methods, and products utilizing a cache coherent switch on chip. The cache coherent switch on chip may utilize Compute Express Link (CXL) interconnect open standard and allow for multi-host access and the sharing of resources. The cache coherent switch on chip provides for resource sharing between components while independent of a system processor, removing the system processor as a bottleneck. Cache coherent switch on chip may further allow for cache coherency between various different components. Thus, for example, memories, accelerators, and/or other components within the disclose systems may each maintain caches, and the systems and techniques described herein allow for cache coherency between the different components of the system with minimal latency.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This patent document claims the benefit and priority of US Provisional patent application Ser. No. 63/223,045 to Shah et al., filed on Jul. 18, 2021, and entitled “Disaggregated servers and virtual resource appliance to compose an application server by allocating and deallocating the components from the pool of volatile memory, persistent memory, solid state drives, input/output devices, artificial intelligence accelerators, graphics processing units, FPGAs and domain specific accelerator components via CXL connected to cache coherent switch SoC and composable management software,” which is hereby incorporated by reference in its entirety for all purposes.

BACKGROUND

(1) As machine learning and other processes become common, datasets continue to grow in size. As the size of datasets increase, the datasets become impractical to store and, thus, processing on the datasets must be efficiently performed to extract useful insight from such datasets.

SUMMARY

(2) Described are methods and systems utilizing cache coherent switch on chip. In a certain embodiment, a system may be disclosed. The system may include a first server device. The first server device may include a first accelerator, a second accelerator, and a first cache coherent switch on chip, communicatively coupled to the first accelerator and the second accelerator via a Compute Express Link (CXL) protocol, where the first cache coherent switch on chip is configured to provide cache coherency between the first accelerator and the second accelerator.

(3) In another embodiment, a method may be disclosed. The method may include receiving, with a cache coherent switch on chip from a network interface card, cache coherent data addressed to a first accelerator, providing, by the cache coherent switch on chip to the first accelerator, the cache coherent data, receiving, with the cache coherent switch on chip from the first accelerator, a bias change, providing, by the cache coherent switch on chip to a processor, the bias change, receiving, with the cache coherent switch on chip from the processor, line resolved data, and providing, by the cache coherent switch on chip to the first accelerator, the line resolved data to cause the first accelerator to write the cache coherent data into a cache coherent memory of the accelerator.

(4) In a further embodiment, a system may be disclosed. The system may include a first Compute Express Link (CXL) device including a CXL interface and a networking component, where the CXL interface is configured to communicate with the networking component over a first software stack via a CXL protocol, and where the CXL protocol includes a L2 layer including a configurable size interframe gap (IFG).

(5) Illustrative, non-exclusive examples of inventive features according to the present disclosure are described herein. These and other examples are described further below with reference to figures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure may best be understood by reference to the following description taken in conjunction with the accompanying drawings, which illustrate various embodiments.

(2) FIG. 1 illustrates a block diagram of an example system, in accordance with some embodiments.

- (3) FIG. 2 illustrates a block diagram of an example cache coherent switch on chip, in accordance with some embodiments.
- (4) FIG. 3 illustrates a block diagram of another example cache coherent switch on chip, in accordance with some embodiments.
- (5) FIGS. 4-10 illustrate block diagrams of example systems, in accordance with some embodiments.
- (6) FIG. 11 illustrates a block diagram of an example cache coherent switch on chip with accelerator, in accordance with some embodiments.
- (7) FIGS. 12-14 illustrate block diagrams of further examples, in accordance with some embodiments.
- (8) FIG. 15 illustrates a block diagram of an example computing system with a cache coherent switch on chip, in accordance with some embodiments.
- (9) FIG. 16 illustrates a block diagram of a networked system, in accordance with some embodiments.
- (10) FIG. 17 illustrates a block diagram of an example cache coherent switch on chip with erasure code accelerator, in accordance with some embodiments.
- (11) FIG. 18 illustrates a block diagram of a system, in accordance with some embodiments.
- (12) FIG. 19 illustrates a software stack, in accordance with some embodiments.
- (13) FIGS. 20A and 20B illustrate IP Security headers, in accordance with some embodiments.
- (14) FIGS. 21A to 32B illustrate various frame formats, in accordance with some embodiments.

DETAILED DESCRIPTION

- (15) In the following description, specific details are set forth to provide illustrative examples of the systems and techniques described herein. The presented concepts may be practiced without some, or all, of these specific details. In other instances, well known process operations have not been described in detail to avoid unnecessarily obscuring the described concepts. While some concepts will be described with the specific examples, it will be understood that these examples are not intended to be limiting.
- (16) For the purposes of this disclosure, certain Figures may include a plurality of similar components. The plurality of such components may be indicated with A, B, C, D, E, F, G, H, . . . N, and/or such indicators to distinguish the individual such components within the Figures. In certain instances, references may be provided to such components without reference to the letter indicators. It is appreciated that, in such instances, disclosure may apply to all such similar components.
- (17) Components described herein are referred to with a three digit ordinal indicator number. In certain instances of this disclosure, certain components may be described herein within a plurality of Figures. In such instances, similar components appearing in a plurality of Figures may include the same final two digits of the three digit ordinal indicator number (e.g., X02).
- (18) Some embodiments of the disclosed systems, apparatus, methods and computer program products are configured for implementing cache coherent switch on chip. As described in further detail below, such a system may be implemented utilizing the Compute Express Link (CXL) interconnect open standard. Such a CXL based cache on chip allows for low latency paths for memory access and coherent caching between devices.
- (19) Utilizing CXL, the currently disclosed cache coherent switch on chip allows for connection of a variety of components connected through a high speed low latency interface. The currently disclosed cache coherent switch on chip allows for multi-host access and the sharing of resources. The cache coherent switch on chip allows for greater utilization of resources, creation of composable virtual servers aligned with workloads, higher efficiency and performance of systems, and flexibility for architecture modifications of systems. The features of the cache coherent switch on chip allows for more efficient utilization of resources and power consumption while providing increased system level performance.

(20) The disclosed cache coherent switch on chip allows for component disaggregation and server composability through system resource sharing without requiring a processor to control such resource sharing and, thus, becoming a bottleneck. As such, system resources may be more fully utilized and resource sharing may optimize component usage within a system, enabling more workloads to be executed. The cache coherent switch on chip also decreases the burden on the system processor, as the system processor is no longer required to handle data and memory transfers and other such tasks.

(21) Furthermore, the disclosed cache coherent switch on chip allows for cache coherency between various different components. Thus, for example, memories, accelerators, and/or other components within the disclosed systems may each maintain caches, and the systems and techniques described herein allow for cache coherency between the different components of the system with minimal latency.

(22) As the size of datasets and the speeds required to process them grow, the value of effective caching and access to such caches becomes ever more valuable. In various embodiments, the systems and techniques may provide for a switch on chip for the caching layer of memory. Thus, cached data, as well as other transient data, may be shared between various devices of a system without requiring CPU involvement. The sharing of cached data or another such transient data may provide for much faster access to such cached data and significantly increase the amount of cached data that may be effectively stored within a system. Accordingly, the systems and techniques provide for switching and sharing of cached data, allowing for data to be accessed at a much faster speed without CPU involvement and for greater optimization of storage of such cached data. Due to CPU involvement no longer being required, a much greater amount of cached data may be shared between various memories, accelerators, graphics cards, and/or other devices.

(23) In various embodiments, a cache hierarchy may be determined and/or utilized by one or more cache coherent switch on chip caches, indicating which caches are prioritized for refreshing and/or reading/writing. In certain embodiments, such caches may be configured to fetch, read, and/or write data according to such hierarchy. Packet flow of data between various components, as well as for caching, may thus be optimized.

(24) FIG. 1 illustrates a block diagram of an example system, in accordance with some embodiments. FIG. 1 illustrates system **100** that includes cache coherent switch on chip **102**, processor **104**, network **170**, accelerators **106**, storage **108**, application specific integrated circuit (ASIC) **110**, persistent memory (PM) **112**, and memory module **114**. Various components of system **100** may be communicatively and/or electrically coupled with a CXL interface **116**, which may be a port. Accordingly, communicative couplings indicated by an interface such as a CXL interface (and/or a PCI or other such interface, as described herein) may each include a corresponding port to establish a signal connection between the two components. Such connections may be indicated by a line with arrows on both ends in the Figures provided herein. Though reference may be made herein to such interfaces, it is appreciated that such references to interfaces may also include the corresponding port of the components (e.g., the ports of the corresponding cache coherent switch on chip). Additionally, other components of system **100** may be communicatively and/or electrically coupled with other interfaces **118**, such as Peripheral Component Interconnect (PCI) and/or other such interfaces. Other such interfaces may be indicated by a line without arrows in the Figures provided herein.

(25) Processor **104** may be any type of processor, such as a central processing unit (CPU) and/or another type of processing circuitry such as a single core or multi-core processor. Processor **104** may be a main processor of an electronic device. For the purposes of this disclosure, “processor,” “CPU,” “microprocessor,” and other such reference to processing circuitry may be interchangeable. Thus, reference to one such component may include reference to other such processing circuitry. In various embodiments, an electronic device or system may include one or a plurality of processors **104**. Each processor may include associated components, such as memory **114B**. Memory **114B**

may, for example, be a memory module, such as a dual in-line memory module, and may provide memory for processor **104**.

(26) Cache coherent switch on chip **102** may be configured to allow for sharing of resources between various components of system **100**, as described herein. Such components may include, for example, accelerators **106A** and **106B**, storage **108** (e.g., smart storage such as harddrives or memories such as solid state drives), ASIC **110**, PM **112**, and memory **114A**. Accelerators **106A** and **106B** may be hardware or software configured to accelerating certain types of workloads and are configured to more efficiently perform such specific workloads. Storage **108** may be harddrives and/or other storage devices. ASIC **110** may be, for example, artificial intelligence ASICs and/or other such ASICs configured to perform specific tasks. PM **112** may be non-volatile low latency memory with densities that are greater than or equal to DRAM, but may have latencies that are greater than DRAM. Memory **114A** may be, similar to memory **114B**, a memory module including random access memory (RAM) and/or another such memory.

(27) In various embodiments, cache coherent switch on chip **102** may be communicatively coupled to one or more such components of system **100** via CXL interface **116**. Cache coherent switch on chip **102** may be configured to allow for sharing of resources between the various such components. In certain embodiments, cache coherent switch on chip **102** may include its own resources, such as its own RAM module, as well as other such resources that are described herein. Such resources may also be shared between the various components. Cache coherent switch on chip **102** may utilize CXL interface **116** to provide low latency paths for memory access and coherent caching (e.g., between processors and/or devices to share memory, memory resources, such as accelerators, and memory expanders). CXL interface **116** may include a plurality of protocols, including protocols for input/output devices (IO), for cache interactions between a host and an associated device, and for memory access to an associated device with a host. For the purposes of this disclosure, reference to a CXL interface or protocol described herein may include any one or more of such protocols. Cache coherent switch on chip **102** may utilize such protocols to provide for resource sharing between a plurality of devices by acting as a switch between the devices.

(28) Typically, all components of a system are controlled via a processor. Thus, component-to-component traffic is controlled by the processor. In such a configuration, the processor, due to limited resources, becomes a bottleneck in component-to-component traffic, limiting the speed of component-to-component traffic. The techniques and systems described, such component-to-component traffic is controlled via cache coherent switch on chip **102**, with CXL interface **116**, generally bypassing processor **104**. As CXL interface **116** allows for an extremely low latency interface between components, processor **104** is no longer a bottleneck and sharing of resources may be performed more quickly and efficiently.

(29) FIG. 2 illustrates a block diagram of an example cache coherent switch on chip, in accordance with some embodiments. FIG. 2 illustrates cache coherent switch on chip **202**. Cache coherent switch on chip **202** includes one or more upstream ports **220** and one or more downstream ports **222**. Each of upstream ports **220** and downstream ports **222** may be configured to support PCI or CXL protocol. As such, upstream ports **220** and downstream ports **222** may be ports configured to support any combination of PCI and/or CXL protocols.

(30) In certain embodiments, one or more upstream ports **220** may be configured to support CXL protocols while one or more downstream ports **222** may be configured to support PCI and CXL protocols. In another embodiment, one or more upstream ports **220** may be configured to support PCI protocols while one or more downstream ports **222** may be configured to support CXL protocols. In a further embodiment, one or more upstream ports **220** may be configured to support PCI protocols while one or more downstream ports **222** may be configured to support PCI protocols. In yet another embodiment, one or more upstream ports **220** may be configured to support CXL protocols while one or more downstream ports **222** may be configured to support

CXL protocols.

(31) Cache coherent switch on chip **202** may include switched fabric circuitry **276** that includes a plurality of nodes and may interconnect a plurality of ports. Switched fabric circuitry **276** may be configured to receive input and/or provide output to the various ports. Accordingly, switched fabric circuitry **276** may be coupled to downstream ports **220**, upstream ports **222**, and/or other ports and/or portions of cache coherent switch on chip **202**. Switched fabric circuitry **276** may be circuitry configured in a switched fabric manner, to allow for inputs and outputs to be interconnected and signals accordingly communicated.

(32) Cache coherent switch on chip **202** may include processing core **274**. Processing core **274** receives electrical signals from ports of cache coherent switch on chip **202** and transforms and/or outputs associated electrical signals to other ports of cache coherent switch on chip **202**. Processing core **274** may be configured to transform signals from a first protocol to a second protocol, and/or may be configured to determine the appropriate port to output signals toward.

(33) FIG. 3 illustrates a block diagram of another example cache coherent switch on chip, in accordance with some embodiments. FIG. 3 illustrates cache coherent switch on chip **302** that includes upstream ports **304** and downstream ports **306**. Furthermore, cache coherent switch on chip **302** may include a plurality of virtual hierarchies **324** (e.g., virtual hierarchies **324A** and **324B**, as well as possibly additional virtual hierarchies) and processor **326**. Each virtual hierarchy **324** may include a combination of PCI and CXL protocols. Any combination of devices described herein may be coupled to upstream ports **304** and/or downstream ports **306**, including memory devices, accelerators, and/or other such devices.

(34) In various embodiments, a cache hierarchy may be determined and/or utilized by cache coherent switch on chip **302**. The cache hierarchy may be, for example, a version of virtual hierarchy **324** and may indicate the priority for the caches of components coupled to cache coherent switch on chip **302**. The cache hierarchy may indicate a priority for refreshing and/or reading/writing the caches of the various components. Such a cache hierarchy may be determined by cache coherent switch on chip **302** based on machine learning according to the techniques described herein and/or may be a preset hierarchy (e.g., a preset hierarchy of which caches of certain components are given priority and/or which components are given priority in utilization of the caches). In certain embodiments, such caches may be configured to fetch, read, and/or write data according to such hierarchy (e.g., higher priority components may be given priority for fetching, reading, and/or writing data to caches, according to the cache hierarchy).

(35) In certain embodiments, one or more of upstream ports **304** and/or downstream ports **306** may include a bridge (e.g., a PCI-to-PCI bridge (PPB)) for coupling to the ports to devices.

Furthermore, cache coherent switch on chip **302** may include one or more virtual bridges (e.g., vPPB) for binding to one or more components coupled to cache coherent switch on chip **302**. In various embodiments, such bridges may additionally include bridges such as SR2MR (Single Root to Multiple Root), SLD2MLD (Single Logical Device to Multi Logical Device), and/or other such legacy bridges to provide for communications with legacy devices.

(36) In certain embodiments, SR2MR bridges may be configured to allow a single root PCIe device to be exposed to multiple host ports. For SR2MR bridges, downstream ports may implement one or a plurality of virtual point-to-point (P2P) bridges. In certain embodiments, one virtual P2P bridge may be utilized for each virtual hierarchy. The SR2MR bridges may be a part of a switch on chip or may be a separate chip communicatively coupled to the switch on chip.

(37) In certain embodiments, SLD2MLD bridges may be configured to allow a CXL standard single logical device to be seen as a multi logical device by the switch domain. Downstream ports implement address translation and enforces the isolation normally performed by multi logical devices. The SLD2MLD bridges may be a part of the switch on chip or may be a part of a separate chip communicatively coupled to the switch on chip.

(38) FIGS. 4-10 illustrate block diagrams of example systems, in accordance with some

embodiments. FIG. 4 illustrates system **400** that includes a plurality of cache coherent switch on chips **402**, CPUs **404**, a plurality of memories **414**, and a plurality of devices **428**. While the embodiment shown in FIG. 4 illustrates a configuration where cache coherent switch on chip **402A** is communicatively coupled (via CXL interface **416**) to CPU **404A** and cache coherent switch on chip **402B** is communicatively coupled to CPU **404B**, in various other embodiments, a single CPU may be coupled to both cache coherent switch on chip **402A** and **402B**. In various embodiments, CPU **404A** and **404B** may be communicatively coupled with interface **418**. One or both of CPUs **404A** and **404B** may be in an active state or one of CPUs **404A** and **404B** may be demoted to a passive state. When in the passive state, the passive CPU may not control downstream devices **428** and, thus, control of such devices **428** may be exclusively by the active CPU.

(39) Cache coherent switch on chips **402A** and **402B** may be communicatively coupled via expansion port **472**. In certain embodiments, cache coherent switch on chips **402** may include processing cores **474**. Expansion port **472** may be a port on cache coherent switch on chips **402** to allow for expansion of processing power of cache coherent switch on chips **402** by, for example, allowing for interconnection of processing cores **474** (e.g., processing cores **474A** and **474B**). Expansion port **472** thus allows for increase in processing power and, in certain embodiments, expansion in the amount of component resources that may be shared. Accordingly, for example, memories **414B**, **414C**, **414E**, and **414F** as well as devices **428A** to **428D** may all be pooled resources for system **400**. Memories **414** may be any type of appropriate memory described herein. One or more memories **414** may form a memory bank for portions of system **400**, such as for one or more cache coherent switch on chips **402**. Devices **428** may be any sort of device of a computing system, such as harddrives, graphics cards, ASICs, I/O devices, and/or other such devices. Furthermore, communicatively and/or electrically coupling together cache coherent switch on chips **402A** and **402B** may provide for greater system redundancy, increasing reliability.

(40) Though the embodiment of FIG. 4 illustrates cache coherent switch on chips **402A** and **402B** being electrically and/or communicatively coupled between expansion port **472**, other embodiments may couple various cache coherent switch on chips with other techniques, such as over a local area network (LAN), over the internet, and/or over another such network.

(41) In certain embodiments, each of cache coherent switch on chip **402A** and **402B** may include their own virtual hierarchies. When coupled as in FIG. 4, the virtual hierarchies of one or both of cache coherent switch on chips **402A** and **402B** may be utilized for switching operations.

(42) FIG. 5 illustrates system **500** that includes cache coherent switch on chips **502A** to **502C**, CPUs **504A** and **504B**, management **530**, and devices **528**. Each of cache coherent switch on chips **502A**, **502B**, and **502C** may include their own individual virtual hierarchies. In certain embodiments, cache coherent switch on chips **502** may include a fabric manager **540** to manage resources connected to the ports (e.g., ports **516**) of cache coherent switch on chips **502**. The fabric manager **540** may connect to higher level management software entities (e.g., management **530**) via Ethernet **568** (as, for example, Redfish over Ethernet) and/or another network or protocol (e.g., PCI protocols). Ethernet **568** may further communicatively and/or electrically couple cache coherent switch on chips **502A**, **502B**, and **502C** and CPUs **504** and devices **528**.

(43) Fabric manager **540** may be configured to allocate and/or deallocate resources attached to the ports of cache coherent switch on chips **502** to applications running on such ports (e.g., to applications running on ASICs coupled to ports of cache coherent switch on chips **502**). Fabric manager **540** may be configured to receive signals (e.g., data) from an upstream port and direct the signal to the appropriate downstream port. Various techniques for receiving and directing such signals (e.g., packet flows) are described herein. Fabric manager **540**, as well as other firmware and/or software may further manage hot plug coupling by devices **528** to downstream CXL ports. Fabric manager **540** may also manage the inventory of various devices coupled to the ports of the respective cache coherent switch on chip **502**.

(44) Fabric manager **540** may be communicatively coupled to management **530** for top level

management of system **500**, including management of the various cache coherent switch on chips **502** described herein. Thus, in various embodiments, management **530** may be, for example, a baseboard management controller and/or another management device or server configured to provide management/orchestration. In various embodiments, management **530** may interface with fabric management **540** to provide for management of the various cache coherent switch on chips (e.g., via a specific fabric management API).

(45) Fabric manager **540** may be implemented within firmware of cache coherent switch on chip **502** (e.g., within the firmware of a microprocessor of cache coherent switch on chip **502**). Such firmware may include a system fabric manager that implements the logic for operations to be performed by switch hardware and other helper functions for implementing the API and a CXL fabric manager for implementing the front-end fabric manager APIs according to the CXL specifications.

(46) In certain embodiments, a CXL single logical device (SLD), such as device **528A**, may be hot-inserted into or hot-removed from cache coherent switch on chip **502B** (e.g., via port **516E**, which may be a PCI and/or CXL protocol port). When such an SLD is first hot-inserted, it is assigned to fabric manager **540B**. Diagnostics may be performed on the newly inserted SLD (e.g., either run as self-diagnostics by device **528A** or run via diagnostics software on the processing core of cache coherent switch on chips **502**). After the SLD has been determined to be ready, it can be assigned to one of the ports (e.g., port **516E**) of cache coherent switch on chip **502B** based on policy (e.g., due to a virtual hierarchy) or via a command (e.g., from software within system **100**).

(47) The assignment may include binding the corresponding downstream PPBs of a cache coherent switch on chip **502** to one of the vPPBs, virtual hierarchies, and host port of cache coherent switch on chip **502**. The managed hot-inserted device **528A** is then presented to the host port (e.g., port **516E**) after its assignment to the respective virtual hierarchy to allocate device **528A**. The host CPU (e.g., the CPU within the respective cache coherent switch on chip **502**) may then discover device **528A** (e.g., via software), load software for device **528A** and begin communicating with device **528A**.

(48) FIG. **6** illustrates system **600**. System **600** may illustrate cache coherent switch on chip **602**. Cache coherent switch on chip **602** may include a plurality of root ports **632** and a plurality of virtual hierarchies **624**. The plurality of root ports **632** may include the ports described herein, as well as, for example, that of internal components within cache coherent switch on chip **602**, such as microprocessors/CPU and/or other components. Each root port **632** may be assigned to downstream CXL protocol resources. Each virtual hierarchy **624** may include a plurality of vPPBs, where certain vPPBs **634** are associated with root ports **632** and other vPPBs **636** are associated with PPBs **638**. Various multi-logical devices (MLDs) **640** may be coupled to downstream ports via certain PPBs **638**.

(49) Cache coherent switch on chip **602** may include a plurality of root ports **632**. Such root ports **632** may include, for example, ports associated with a processing core of cache coherent switch on chip **602** as well as external devices. Root ports **632** may be assigned to downstream CXL resources, including embedded accelerators within system **600**. Fabric manager **640** may include a processor (e.g., an ARM processor or another type of processor) and such a processor may be a part of one or more virtual hierarchies **624**. Various downstream PPB ports **638** may be communicatively coupled to MLDs **640**. The assignment of MLDs **640**, as well as other components such as SLDs, memories, accelerators, and other such components, to certain PPBs **638** and vPPBs **636** may be controlled by fabric manager **640**. Thus, fabric manager **640** may detect that a component has been coupled to a port of cache coherent switch on chip **602** and accordingly assign the component to the appropriate virtual hierarchy **624** (e.g., based on the detected type of the component). Furthermore, the appropriate PPB **638** and/or the vPPB **636** may be assigned to the component. In certain embodiments, such assignment may be based on the detected type of the component and on virtual hierarchy **624**.

(50) FIG. 7 illustrates system **700** that includes cache coherent switch on chip **702**. As shown in FIG. 7, MLDs **740A** and **740B** are coupled to PPBs **738A** and **738B**, respectively. MLDs **740A** and **740B** include memories **714A** and **714B**, respectively, and are thus utilized as memory expansion. Coupling of memories **714A** and **714B** to system **700** allows for an increase in the amount of memory of system **700** (e.g., system **700** may be, for example, a single socket server).

(51) In various embodiments, the amount of memory attached to a socket is limited by the number of channels that the socket supports. In certain situations, in a data-centric environment, an entire operating data set may not fit in a server's available memory, resulting in poor performance and increased latency when processing the data. Cache coherent switch on chip **702** addresses this problem by allowing for low-latency memory expansion due to memories **714A** and **714B** via the ports of cache coherent switch on chip **702**, increasing the amount of memory available to a host CPU (beyond what could be connected directly to the CPU). Memories **714** may be DDR4, DDR5, future DDR, DRAM, PM, NVMe, Low-Power Double Data Rate (LPDDR), and/or other such appropriate memory drives which may be expanded via CXL protocol through cache coherent switch on chip **702**.

(52) Such an ability of cache coherent switch on chip **702** is particularly beneficial in providing cost and performance advantages for memory intensive applications that would otherwise require a computing device with a large memory footprint or result in poor performance in a less expensive computing device with limited memory.

(53) FIG. 8 illustrates system **800** that includes a plurality of servers **842A** and **842B**. Each server **842** may include its own cache coherent switch on chip **802**, a plurality of memories **814** communicatively coupled to each cache coherent switch on chip **802**, and a microprocessor **804** communicatively coupled to each cache coherent switch on chip **802**. Cache coherent switch on chip **802A** and **802B** may be communicatively coupled via fabric switch/bus **844**. In various embodiments, fabric switch/bus **844** may be, for example, a switch fabric, a bus bar, and/or another such technique for communicating signals between different server devices.

(54) As illustrated in FIG. 8, memories may be pooled between different microprocessors **804**. Such memories may include memories **814** communicatively coupled to cache coherent switch on chips **802** and/or memory that is socket connected to various microprocessors **804**. Thus, cache coherent switch on chips **802** may allow for pooling of memory and other resources (e.g., AI, ASICs, GPUs, SNICs, NVMe, storage, and/or other such resources) between servers **842** that are communicatively coupled via switch fabric/bus **844**. As signals communicated between switch fabric/bus **844** may be similar to that of signals communicated within a single server device, cache coherent switch on chips **802** may allow for sharing of such resources in a similar manner to that described herein. In various embodiments, a plurality (two or more) of servers **842** may, accordingly, pool memory resources such as DRAM, PM, and/or other such memories. Such resources may be shared over fabric switches for memory pooling inside a server, between servers within a server rack, between various servers and racks within a data center, and/or between data centers. In a further embodiment, messages may be passed between components in a manner similar to that of the sharing of resources. Such techniques allow for reduction in the communication of messages between various components, increasing the performance of, for example, AI or ML workloads on processors.

(55) In various embodiments, cache coherent switch on chips **802** may provide compression and/or decompression ability to conserve persistent memory as well as crypto ability to provide added security between transactions into and out of persistent memory.

(56) In certain embodiments, a prefetched buffer scheme may be utilized at the memory source. Accordingly, in various embodiments, cache coherent switch on chips **802** may include memory prefetchers **878**. Memory prefetchers **878** may be an intelligent algorithm run by the processing core of the cache coherent switch on chips **802**. Memory prefetchers **878** may be an artificial intelligence (AI) or machine learning (ML) prefetcher configured to predict the addresses of future

accesses to memories based on past access patterns by the hosts, and prefetch data from such memories for those addresses to store in DRAM buffers to reduce the latency of future accesses by the host applications. In certain embodiments, accelerators communicatively coupled to cache coherent switch on chip **802** may also be configured to provide prefetching when pooling resources via cache coherent switch on chips **802** between servers **842A** and **842B**.

(57) In certain embodiments, disaggregated servers **842** may pool memory and/or other resources across a midplane (e.g., bus **844**). Thus, for example, in a chassis or blade server, a large shared pool of memory on memory cards/blades is available to be used by server cards/blades (that could be lightweight servers, aka thin servers, with a minimal amount of their own memory connected to the CPU socket). Such memory pooling may provide cost and/or power consumption advantages by reducing the amount of unused memory and/or other resources in data center servers, as memory/resource pooling allows for greater flexibility and, thus, a lower requirement for fixed resources. Servers may also be more flexibly configured due to the advantages of resource sharing.

(58) In a certain use case, current typical server systems may include 512 gigabyte (GB) or so of volatile memory in cloud service provider infrastructure. A portion of this memory is typically stranded due to lower memory utilization for all the applications. Additionally, certain cloud environments include highly memory intensive applications that require more than 512 GB of memory. Currently, for example, platforms allocate all the servers with 512 GB memory due to simplicity, stranding the memory resources in the majority of the servers in order to have enough capacity for edge use cases. The currently disclosed cache coherent switch on chips addresses this memory stranding problem by allowing for the sharing of CXL protocol persistent memory both inside the server system and to outside servers connected via a network.

(59) FIG. **9** illustrates system **900** that includes server **942**, switch fabric/bus **944**, and memory appliance **946**. Memory appliance **946** may be a shared or expansion memory for server **942**. System **900** allows for memory **914A** of cache coherent switch on chip **902A** to be declared as a cache buffer for persistent memory ports (e.g., ports coupled to switch fabric/bus **944** and, thus, memory **914C** of memory appliance **946**). Utilizing memory **914A** as a read/write buffer hides the access time of utilizing memory appliance **946** and, thus, memory **914C**.

(60) In various embodiments, there may be both write and read flows for memory **914A**. In a write flow, microprocessor **904** may indicate that writes on memory **914A** are steered to a DRAM buffer port of cache coherent switch on chip **902A**. For such writes, cache coherent switch on chip **902A** may check to ensure that memory **914C** is configured to provide buffer write/read commands to memory **914A**, allowing for memory **914A** to be used as a buffer for memory **914C**. Thus, memory **914C** is updated so that the buffer write/read address of memory **914C** refers to that of memory **914A**. Memory **914A** may then be accordingly utilized as a buffer for memory **914C**, avoiding the increase in access time of utilizing memory appliance **946**.

(61) In certain embodiments, for a read flow, microprocessor **904** may first query the buffer port of memory **914A** for the wanted data. If such data is present within the buffer of memory **914A**, the data may be provided to microprocessor **904**. If memory **914A** does not include such data, memory **914C** may be queried and the requested data may be provided from memory **914C** over switch fabric/bus **944**.

(62) In certain embodiments, the cache buffers of memory **914A** include AI/ML prefetch algorithms. The algorithm is configured to predict the next set of addresses (expected to be fetched by the applications) and configures a direct memory access (DMA) engine to prefetch those addresses and store the data in read/write buffers, to be ready to be read by the applications. In certain embodiments, cache coherent switch on chip **902A** is configured to keep statistics of hit ratios for each line that was prefetched to provide feedback to the algorithm for continuous improvement (e.g., to determine which prefetched data has been utilized).

(63) In certain embodiments, cache coherent switch on chip **902A** may provide instructions for operation of the memory prefetcher. Thus, cache coherent switch on chip **902A** may be configured

to determine data to be prefetched (e.g., based on the AI/ML prefetch algorithm) and provide instructions (via switch fabric/bus **944**) to memory **914C** to provide such prefetched data to memory **914A** (via switch fabric/bus **944**) for caching. Memory **914C** may accordingly provide such data for buffering by memory **914A**.

(64) In certain embodiments, each upstream port of cache coherent switch on chip **902A** is configured to determine whether a cache buffer port is assigned for the respective upstream port. If a cache buffer port is assigned, a further determination may be made as to which downstream port is assigned as the cache buffer port. Incoming traffic may then be accordingly provided to the assigned downstream port for cache buffer purposes.

(65) In various embodiments, caching may be performed by memory of the switch on chip and/or memory attached to the ports of the switch on chip. Various, cache coherent switch on chip **914A** may determine whether requested data is within the cache and retrieve such data if it is present within the cache. If the data is not within the cache, a request may be provided to the coupled persistent memory for the data and the data may be accordingly provided. In certain embodiments, write requests may be provided to both the cache and the persistent memory.

(66) FIG. **10A** illustrates system **1000** that includes servers **1000A** and **1000B**. Each server **1000A/B** includes a cache coherent switch on chip **1002**, each cache coherent switch on chip **1002** communicatively/electrically coupled to CPU **1004**, accelerator **1006**, storage **1008**, ASIC **1010**, PM **1012**, memory **1014**, and network interface card (NIC) **1080**. Each accelerator **1006** may include respective memory **1046**, which may include its own cache coherent and non-cache coherent storage. Cache coherent switch on chips **1002A** and **1002B** may be communicatively coupled via network/bus **1044** via NICs **1080A** and **1080B**. FIG. **10A** may illustrate a configuration where a cache coherent switch on chip of a first server may bridge over Ethernet to another cache coherent switch on chip of a second server and allow for the sending and receiving (and, thus, reading and writing) of cache coherent traffic directly between NIC **1080** and accelerator **1006**'s cache coherent memory, via cache coherent switch on chip **1002**.

(67) In various embodiments, cache coherent switch on chips **1002A** and **1002B** may be communicatively coupled via an Ethernet connection (e.g., via network **1044**). As such, cache coherent switch on chips **1002** may communicate via CXL protocol through Ethernet to allow for resource pooling and/or sharing (e.g., of memory, accelerators, and/or other devices) between different devices, server racks, and/or data centers.

(68) In various embodiments, commands received from a host via a CXL protocol port of cache coherent switch on chips **1002** are received and terminated inside the respective cache coherent switch on chips **1002** at the CXL protocol port. Cache coherent switch on chip **1002** may then provide a corresponding command tunneled within the payload of Ethernet frames that are communicated over network **1044**. Thus, cache coherent switch on chip **1002** includes a bridging function that is configured to terminate all the read and write commands (e.g., persistent memory flush commands) inside cache coherent switch on chip **1002** and provide corresponding commands over Ethernet.

(69) NICs **1080** may be configured to allow for cache coherent switch on chips **1002s** to communicate via network/bus **1044**. In certain embodiments, cache coherent switch on chips **1002** may be provided for data flow between accelerators **1006** and NICs **1080** (which may be a Smart NIC) so that NICs **1080** may write directly into accelerator **1006**'s cache coherent memory. Such data flow allows for sending and/or receiving of cache coherent traffic over network **1044** by accelerators **1006**.

(70) The configuration of system **1000** allows for data to be communicated between components within servers **1000A** and **1000B** as well as between servers **1000A** and **1000B** without needing to be controlled by CPUs **1004**. Furthermore, the components of system **1000** are decoupled from each other, with traffic controlled by respective cache coherent switch on chips **1002**.

(71) In a certain embodiments, system **1000** may be configured so that cache coherent traffic stays

within respective servers **1000A** and **1000B**. Cache coherency within each server **1000A/B** is resolved by respective CPU **1004**. Cache coherent switch on chips **1002** may provide accelerator traffic over network **1044**, but in certain such embodiments, such accelerator traffic may be non-cache coherent traffic. The cache coherent traffic is thus never exposed to network **1044**.

(72) In certain embodiments, (e.g., with processing core **474** within a cache coherent switch on chip, as described in FIG. 4), cache coherent switch on chips **1002** may be configured to resolve cache coherent traffic among accelerators **1006**, as well as resolve cache coherency within CPU **1004**. Thus, for example, cache coherent switch on chips **1002** may resolve symmetric coherency between two processing domains based on CXL protocol (e.g., allow for coherency between accelerator **1006** and CPU **1004**). In various embodiments, the processing core within cache coherent switch on chip may receive and provide cache coherent traffic between the various components of system **1000**, including accelerator **1006**, CPU **1004**, as well as other components. Thus, for example, all cache coherent traffic may be provided to cache coherent switch on chip **1002** and cache coherent switch on chip **1002** may then provide corresponding cache coherent traffic to respective target components. In such a configuration, CPU **1004** is no longer in charge of cache coherency, or the sole communicator of such data thereof. Instead, cache coherent switch on chip **1002** may resolve cache coherency between accelerator **1006** and any number of components within system **1000** (e.g., by determining that data received is cache coherency data and providing such coherency data to the respective components). Thus, for example, cache coherent switch on chip **1002** may include instructions to provide cache coherency data to one or more components for any received data. Such a configuration may reduce the cache coherency traffic between accelerators and CPUs, as well as other components within system **1000**, increasing the performance of accelerator dominated ML/AI workloads by alleviating the bottleneck of CPUs. Such a configuration may also allow for cache coherency between different accelerators of multiple different systems, which are managed by their respective cache coherent switch on chips, increasing the total number of accelerators that are cache coherent in a given system and, thus, allow for a large batch of coupled accelerators for increased performance.

(73) In a further embodiment of providing/receiving cache coherent traffic to accelerator **1006** over network **1044**, NIC **1080** may indicate that it is providing cache coherent traffic to accelerator **1006**. Upon receipt of such traffic, accelerator **1006** may provide the bias change of the coherent memory line to CPU **1004** (via cache coherent switch on chip **1002**). Upon receipt, CPU **1004** may then provide snoop requests to all components (e.g., components snooping for cache coherency) within its respective server (e.g., that of server **1000A** or **1000B**) to provide for cache coherency within all components of the respective server. Once the cache line is resolved, CPU **1004** provides a line resolved message to the requesting accelerator **1006**. Upon receipt of this message, accelerator **1006** may write the received traffic from NIC **1080** into the cache coherent portion of the respective memory **1046** of accelerator **1006** and, accordingly, coherency may be achieved within all components of the respective server.

(74) Typically, accelerator to accelerator traffic within a system is provided via a proprietary switch. Cache coherent switch on chip **1002** allows for the elimination of such a proprietary switch while providing for accelerator to accelerator traffic. Accordingly, CXL protocol data may be provided from a first accelerator **1006A** to a cache coherent switch on chip **1002**, to CPU **1004A**, and then communicated to a second accelerator **1006B** to provide for cache coherency between the accelerators of server **1000A**.

(75) In various embodiments, CPU **1004** may include a home agent configured to resolve coherent traffic. Cache coherent traffic may be resolved by the home agent of CPU **1004**. However, cache coherency may also be resolved within a processing core (e.g., a processing core such as processing core **474** of cache coherent switch on chip) of the cache coherent switch on chip, removing CPU **1004** as a bottleneck. Accordingly, such coherent traffic may be provided by one of accelerator **1006A** and received by cache coherent switch on chip **1002A**. The processing core of cache

coherent switch on chip **1002A** may then provide such coherent traffic to the other accelerators of the coherent group that are communicatively coupled to cache coherent switch on chip **1002A**, such as accelerator **1006B**, as well as other accelerators (e.g., communicatively coupled via network/bus **1044**).

(76) In a typical system, when data arrives from a network, typical data flows include network to processor, processor to storage, storage to processor, and processor to accelerator. As the volume of data grows, the processor becomes a bottleneck in this type of circular cycle of data transfer.

(77) Cache coherent switch on chip **1002** allows for data to flow through to its ultimate destination while bypassing any CPU bottleneck. Thus, cache coherent switch on chip **1002** allows for data transfer between various ports, such as between two downstream ports. Components that are coupled to cache coherent switch on chip **1002** may, accordingly, more easily transfer data between each other and bypass CPU bottlenecks. Such transfers may be of the CXL protocol format.

(78) For data transfers between accelerators and storage devices allocated to a root port of a microprocessor of cache coherent switch on chip **1002**, the transfers may be cache coherent (e.g., controlled by the microprocessor of cache coherent switch on chip **1002**), removing the need for cache coherency to be resolved by CPU **1004**. Such a configuration provides for bandwidth and latency advantages as CPU **1004** may be bypassed and may be especially beneficial for neural networks, cryptocurrency, and/or other such systems where accelerators, ASICs, and/or other devices are primarily used (e.g., during training or mining).

(79) In a first example, NIC **1080** may receive cache coherent traffic from network/bus **1044**. The data may be accordingly provided to cache coherent switch on chip **1002** and provided to memory **1014**. Memory **1014** may provide such cache coherent data to accelerator **1006** as well as to storage **1008**. Thus, accelerator **1004**, memory **1014**, and storage **1008** may each include such coherent data. In various embodiments, accelerators **1006** may be a part of the virtual hierarchy of cache coherent switch on chip **1002** to allow for cache coherency between memory **1014** and accelerator **1006**.

(80) Each cache coherent switch on chip **1002** may be communicatively/electrically coupled with one or more of a plurality of accelerators **1006**. As each cache coherent switch on chip **1002** may be communicatively/electrically coupled to one or more other cache coherent switch on chip **1002**, the number of accelerators available to each of the communicatively/electrically coupled cache coherent switch on chips **1002** may be accordingly expanded across a network to encompass accelerators that are coupled to the plurality of cache coherent switch on chips **1002**. Various, cache coherent switch on chip **1002** may provide for such pooling regardless of whether the respective accelerator is assigned to CPU **1004** or a microprocessor of the cache coherent switch on chip **1002** (allowing for operation of the accelerator via cache coherent switch on chip **1002**).

(81) Thus, cache coherent switch on chip **1002** allows for creating and managing a pool of CXL protocol attached accelerators or other resources distributed across one or more cache coherent switch on chips **1002**. In various embodiments, each cluster of communicatively coupled cache coherent switch on chips **1002** may include their own respective virtual hierarchies and cluster of resources. Resources within each cluster may communicate between each other accordingly as if all are connected to the same switch.

(82) Resources within the pool (such as accelerators) may be allocated/deallocated to any application server inside a rack, aisle, data center, and/or any portion of networked data centers communicatively coupled via CXL protocol (including via CXL protocol over Ethernet or other networks). Applications servers may thus be provided with direct access to all accelerators within a cluster, removing all data transformations that are required in typical architecture (e.g., from CUDA code to RDMA protocol packets and back).

(83) In certain embodiments, traffic passing through a first cache coherent switch on chip may be mirrored on a second cache coherent switch on chip. The mirrored traffic may then be utilized for, for example, analysis of traffic that is provided through the first cache coherent switch on chip.

(84) FIG. 10B illustrates formats of read packet **2000A**, read response packet **2000B**, write packet **2000C**, and write acknowledgment packet **2000D**. Such packets may be used for providing resource pooling (e.g., via bridging) and persistent memory functions over Ethernet. Various, each of packets **2000** may include preamble **2002**, DA **2004**, SA **2006**, type **2008**, command **2010**, address **2012**, and CRC **2016**. Read packet **2000A** and write acknowledgment packet **2000D** may include PAD **2014**. Read response packet **2000B** may include read data **2018** and write packet **2000C** may include write data **2020**. The size of each portion of data may be indicated within FIG. 10B.

(85) For read packet **2000A**, command **2010** may include a command indicating “PM read” with length data of the packet and the intended address. For read response packet **2000B**, command **2010** may indicate “PM response” with the intended address and the read data. CRC **2016** may indicate the full Ethernet frame. Address **2012** may correspond to the persistent memory's address.

(86) For write packet **2000C**, command **2010** may indicate a “PM write” with length data of the packet, the intended address, and the write data. For write acknowledgement packet **2000D**, command **2010** may indicate a “PM write acknowledgement” and the intended address.

(87) In various embodiments, compression and/or decompression may be utilized and, based on the packets, the same compression and/or decompression algorithm may be utilized for both the read initiator and the target. Compressed data may be inflated at the source and written within cache.

(88) FIG. 11 illustrates a block diagram of an example cache coherent switch on chip with accelerator, in accordance with some embodiments. FIG. 11 illustrates system **1100** that includes cache coherent switch on chip **1102**, CPU **1104** with memory **1114B**, and NIC **1180**. Cache coherent switch on chip **1102** includes fabric **1148** and compression and security module (CSM) **1150**. CSM **1150** allows for cache coherent switch on chip **1102** to perform compression and decompression for data received. Such a configuration provides significant advantages over conventional techniques, which typically include separate dedicated compression/decompression hardware that would require multiple data communication steps through the CPU to provide for compression and/or decompression and communication of such compressed and/or decompressed data.

(89) In certain embodiments, after data arrives within cache coherent switch on chip **1102** from the network (e.g., via NIC **1180**), the data is provided to CSM **1150** to be decrypted and/or decompressed. Once the data is decrypted and/or decompressed, such data is then provided to other components (e.g., target components of the data) through one or more ports of cache coherent switch on chip **1102**. Additionally, when data is provided to cache coherent switch on chip **1102** to be provided to the network via NIC **1180**, CSM **1150** may first encrypt and/or compress such data before providing such data to NIC **1180** (and, thus, the network).

(90) FIGS. 12-14 illustrate block diagrams of further examples, in accordance with some embodiments. FIG. 12 illustrates system **1200** that includes a plurality of servers **1242**. Server **1242A** and **1242B** are communicatively coupled via switch **1252** (e.g., cache coherent switch on chips **1202A** and **1202B** of servers **1242A** and **1242B**, respectively, are communicatively coupled via switch **1252**). In various embodiments, multiple such servers may be communicatively coupled via fabric switch. Coupling in such a manner may allow for such communicatively coupled servers (e.g., servers **1242A** and **1242B**) to pool resources such as CXL protocol or CPU socket attached memory, accelerators, and/or other such resources over fabric, increasing the amount of resources available to a system and increasing flexibility. In various embodiments, such resources may be pooled via software controlled, driver, or driver-less techniques.

(91) In a certain instance, server **1242B** may wish to share one or more of memories **1214F-J** with server **1242A**. A driver running within server **1242B** may pin such memory through a registration routine and may provide an access key to server **1242A** for access to the respective memory and configures the respective cache coherent switch on chip **1202B** for access via the key. Cache coherent switch on chip **1202A** of server **1242A** may then access the shared memory via CXL

protocol memory commands. In certain embodiments, such CXL protocol memory commands may include read/write instructions and the key. Receiving such commands, cache coherent switch on chip **1202B** may then perform the appropriate action (e.g., providing the read response for read commands or providing a write acknowledgement for write commands).

(92) In another instance, server **1242B** may share read/write caches with server **1242A**. When recalling cached data, server **1242A** may first check if the data is available locally. If the data is not available locally, a request for cached data is provided to server **1242B**. Server **1242B** may then provide the requested cached data either from a cache within memories **1214F-I** of server **1242B** or from memory **1214J** communicatively coupled to microprocessor **1204B**.

(93) In other embodiments, two or more servers may be a part of the system. A local server may determine that requested data is not within its own buffer and may then communicate requests for the buffer data to each of the various servers. The various servers may provide erasure code, accordingly to the techniques described herein (e.g., within FIG. **17**). The servers receiving the request may each determine whether its own caches include the requested data. Servers that include the data may then provide read responses to the requesting server and the requesting server may then receive erasure code data and replace the missing data blocks. Servers that do not include the data may provide read requests to corresponding memory, update the corresponding caches, and provide the data blocks to the requesting server. The requesting server may then reconstruct such data.

(94) FIGS. **13A** and **13B** illustrate system **1300** that includes downstream bridges for supporting legacy devices. Systems **1300** of FIGS. **13A** and **13B** include cache coherent switch on chip **1302** and bridge **1354**. Bridge **1354** may be, for example, a single root to multi root (SR2MR) or single logical device to multiple logical device (SLD2MLD) bridge. Bridge **1354** may be configured to expose a single device (e.g., device **1328**) to multiple host ports.

(95) In the embodiment of FIG. **13A**, bridge **1354A** may be a SR2MR bridge. In various embodiments, port **1316** may be communicatively coupled to bridge **1354A** via a PCI protocol. Bridge **1354A** may be accordingly communicatively coupled to device **1328** via the PCI protocol. Bridge **1354A** may be implemented within cache coherent switch on chip **1302** or as a separate chip.

(96) Bridge **1354A** may include a plurality of virtual function assignments **1396A-C**. Port **1316** may be coupled to device **1328** via bridge **1354A**. Port **1316** may include a plurality of point-to-point (P2P) bridges **1386A-D**. Each virtual function **1396** may be associated with a corresponding P2P bridge **1386**. Each virtual function **1396** may include address remap logic. In certain embodiments, port **1316** may implement physical function assignment logic to control processor **1398**. Due to the matched virtual functions **1396** of bridge **1354A** to P2P bridges **1386** of port **1316**, device **1328** may be associated with a plurality of roots (e.g., multi-roots). The configuration of system **1300A** may be utilized for single root devices and may provide for the implementation of multi-root devices while providing the security and isolation of separate virtual hierarchies.

(97) In the embodiment of FIG. **13B**, bridge **1354B** may be a SLD2MLD bridge. Bridge **1354B** may be implemented within cache coherent switch on chip **1302** or as a separate chip. Bridge **1354B** may be communicatively coupled to PPB **1338** and, accordingly, vPPBs **1336**. Bridge **1354B** may provide a plurality of address remaps **1356A/B** as well as provide for assignment logic such as for interrupts and resets with **1356C**. Thus, single logic device **1328** coupled to bridge **1354B** may be virtualized into a multi-logic device. A single logic device **1328** may be accordingly associated with a plurality of vPPBs **1336** and available as a resource and/or utilize resources from a plurality of other devices communicatively coupled to cache coherent switch on chip **1302**. Utilizing the configuration of system **1300B**, a single logic device may be shared and become, effectively, a multi-logic device and obtain the security and isolation benefits of a multi-logic device with a plurality of virtual hierarchies.

(98) FIG. **14** illustrates system **1400** with cache coherent switch on chip **1402** with fabric **1448** of

cache coherent switch on chip **1402** coupled to chiplets **1464**. In certain embodiments, chiplet **1464** may be a memory controller chiplet that increases the efficiency and reduces the latency of memory. In other embodiments, chiplets **1464** may be other types of chiplets, such as AI inference engines, FPGAs, GPU accelerators, edge computing devices, and/or other such devices.

(99) FIG. **15** illustrates a block diagram of an example computing system with a cache coherent switch on chip, in accordance with some embodiments. FIG. **15** illustrates system **1500** that includes cache coherent switch on chip **1502**. Cache coherent switch on chip **1502** may be communicatively coupled to a resource pool. The resource pool may include a plurality of CPUs **1504A-N**, devices **1528**, accelerator **1506**, memory **1514**, storage **1508**, processor **1504**, and ASIC **1510**. Such communicative coupling may be via a CXL protocol. Such resource pools may be within a server, within a data center, and/or communicatively coupled via Ethernet, the Internet, and/or another data connection (e.g., Bluetooth or satellite Internet).

(100) As described herein, cache coherent switch on chip **1502** may be configured to assign one or more resources from the resource pool to applications on demand. When the application no longer requires the assigned resources, the resources may be reallocated available for other applications.

(101) FIG. **16** illustrates a block diagram of a networked system, in accordance with some embodiments. Networked system **1600** may include a plurality of server racks **1666**. Server racks **1666A** and **1666B** may be communicatively coupled via Ethernet **1668A** and server racks **1666C** and **1666D** may be communicatively coupled via Ethernet **1668B**. Ethernet **1668A** and **1668B** may be communicatively coupled via Internet **1670**. Accordingly, server racks **1666A-D** may all be communicatively coupled with each other.

(102) Each of server racks **1666A-D** may include their respective cache coherent switch on chips. Resource clusters may be created from devices communicatively coupled to the respective cache coherent switch on chips within a server rack (e.g., within one of server racks **1666A** to **D**), from devices communicatively coupled via Ethernet **1668**, from devices communicatively coupled via Internet **1670**, and/or communicatively coupled via another technique. Accordingly, the cache coherent switch on chip disclosed herein allows for the creation of any resource cluster within a system, within a server rack, and across the server racks, creating completely fungible resources connected via a high speed CXL network or CXL protocol over fabric.

(103) FIG. **17** illustrates a block diagram of an example cache coherent switch on chip with erasure code accelerator, in accordance with some embodiments. FIG. **17** illustrates cache coherent switch on chip **1702** with ports **1720/1722**, fabric **1776**, erasure code accelerator **1782**, and processor **1726**.

(104) Erasure code accelerator **1782** may provide redundancy for data stored in persistent memory, non-volatile memory, random access memory, and/or other such memory communicatively coupled to cache coherent switch on chip **1702** or across a network that cache coherent switch on chip **1702** is communicatively coupled to with other cache coherent switch on chips.

(105) Thus, erasure code accelerator **1782** may be communicatively coupled to processor **1726** and/or to memory or storage communicatively coupled to ports **1720/1722**. In situations where erasure code accelerator **1782** is communicatively coupled to processor **1726**, erasure code accelerator **1782** may perform read/write requests addressed to processor **1726**. Erasure code accelerator **1782** thus stripes data across one or more non-volatile memory on writes and reconstructs data from such memory during reads. In the event of a non-volatile memory failure, erasure code accelerator **1782** may support reconstruction of any lost data.

(106) In certain embodiments, cache coherent switch on chip **1702** may receive a write data flow. For a write data flow received by cache coherent switch on chip **1702**, a check may be performed to determine whether the write data is assigned a virtual end point (e.g., a memory or I/O device) in a virtual hierarchy. If the write is for the virtual end point, fabric **1776** may provide the data to processor **1726**. Processor **1726** may then provide the write request to erasure code accelerator **1782**, identifying the port associated with the request and the erasure code technique for use. Data

may then read from various CXL protocol ports of cache coherent switch on chip **1702**, allowing for erasure coding to be accordingly performed by erasure code accelerator **1782** by modifying the data and recalculating the erasure coded data. The modified erasure coded data is then written to the respective CXL port (e.g., the ports where the data is read from the various CXL protocol ports). Such a technique may conserve processing resources by offloading erasure coding to erasure code accelerator **1782**.

(107) Erasure code accelerator **1782** may also provide a read data flow. In a certain embodiment, ingress logic (e.g., for a read request from a port of cache coherent switch on chip **1702**) determines whether the read data flow has erasure code implemented. If erasure code has been implemented, the read request may be provided to processor **1726**. Processor **1726** may then provide the read request to erasure code accelerator **1782**. The read request may identify the port (and, thus, the device communicatively coupled to the port) where the read request was received. The requested read data may then read from various CXL protocol ports of cache coherent switch on chip **1702**, allowing for erasure coding to be accordingly performed by erasure code accelerator **1782** to prepare new erasure coded data. The erasure coded data is then provided back to the respective requesting CXL port.

(108) The various accelerators of cache coherent switch on chip **1702** (e.g., compression, security, erasure coding, and/or other such accelerators) and processor **1726** of cache coherent switch on chip **1702** may be utilized for provisioning of computational storage services (CSSes) to applications running on host CPUs (e.g., CPUs of the greater system containing cache coherent switch on chip **1702**). For example, processor **1726** and CSM modules may serve as computational storage processors (CSPs) to provide CSSes to attached hosts. Processor **1726** may also be utilized as the host in computational storage use cases, orchestrating data movement and running of CSSes. In certain embodiments, processor **1726** may offload batch processing of CSS commands from the host CPUs.

(109) FIG. **18** illustrates a block diagram of a system, in accordance with some embodiments. FIG. **18** illustrates a system that includes CXL to Ethernet (CXL2Eth) Bridge **1802**, Ethernet Connected Memory **1824**, and host **1814**. CXL2Eth Bridge **1802** includes direct memory access (DMA) **1806**, CXL2Eth module **1808**, and CXL IP **1810**. CXL2Eth module **1808** and CXL IP **1810** may be communicatively coupled via CXL memory **1804** (a direct CXL memory connection) or via DMA **1806** through CXL2Eth **1808** providing a CXL.memory format data **1820**, which is then converted into CXL.io (input/output) format data **1818**.

(110) CXL IP **1810** may be communicatively coupled to host **1814** via CXL format communications **1812**. Host **1814** may include host memory **1816** and may be a host device as described herein. Host **1814** may access Ethernet Connected Memory **1824** via CXL2Eth Bridge **1802**.

(111) CXL2Eth **1808** may be communicatively coupled to Ethernet Connected Memory **1824** via Ethernet **1822**. In certain embodiments, CXL2Eth **1808** may be communicatively coupled to memory controller **1826** of Ethernet Connection Memory **1824**. Memory controller **1826** may provide access to memory **1828** of Ethernet Connected Memory **1824** (e.g., for host **1814**), according to the techniques described herein.

(112) FIG. **19** illustrates a software stack, in accordance with some embodiments. FIG. **19** illustrates various configurations of software stacks for CXL interfaces to SerDes **1902**. Thus, for example, CXL **1910A-E** may interface with SerDes **1902** through the techniques described herein. Such interfaces may be via ERT (Elastics.cloud Reliable Transport) **1908**, which may be a software transport technique providing for CXL **1910A** (which may be CXL 3.0 specification compliant) to couple to an off-site SerDes **1902** via Ethernet Layer 2 (L2 or Data Link Layer) **1904A**. Such techniques may be according to the techniques described herein and may allow for off-site utilization of resources for CXL **1910A** to interface with SerDes **1902** and the associated resource.

(113) Additionally, FIG. **19** includes CXL 3.0 communications **1906**, ROCEV1 (RDMA over

Converged Ethernet version 1) **1916**, ROCEV3 (RDMA over Converged Ethernet version 3) **1928**, Ethernet L2 **1904B**, Modified L2 **1914** (e.g., a modified Data Link Layer that may be utilized not over Ethernet), Media Access Control Security (MACSec) **1912**, multi-protocol label switching (MPLS) **1918**, Internet Protocol (IP) **1920**, IP Security (IPSec) **1922**, Transmission Control Protocol (TCP) **1924**, Secure Sockets Layer (SSL) **1930**, and User Datagram Protocol (UDP) **1926**. Modified L2 **1914** may include tags in the preamble phase and/or a shorter interframe gap (IFG). CXL **1910A** and **1910C-E** may be proprietary formats according to CXL and/or IEEE standards. ROCEV3 **1928** may include select acknowledgements (SACK).

(114) Various, CXL **1910** may communicate with SerDes **1902** through various software stacks as described within FIG. **19**. Thus, for example, CXL **1910D** may communicate over SSL **1930** over TCP **1924** over IPSec **1922** over IP **1920** over MPLS **1918** and over Ethernet L2 **1904B** or Modified L2 **1914**. CXL **1910D** may alternatively communicate over TCP **1924** over IPSec **1922** over IP **1920** over MPLS **1918** and over Ethernet L2 **1904B** or Modified L2 **1914**. CXL **1910E** may communicate over ROCEV3 **1928** over UDP **1926** over IPSec **1922** over IP **1920** over MPLS **1918** and over Ethernet L2 **1904B** or Modified L2 **1914**. CXL **1910C** may communicate over ROCEV1 **1916** and over Ethernet L2 **1904B** or Modified L2 **1914**. CXL **1910C** may communicate over ROCEV1 **1916** over Ethernet L2 **1904B** or Modified L2 **1914** and over MACSec **1912**. CXL **1910B** may be CXL 3.0 specification compliant and may communicate via CXL 3.0 specification with SerDes **1902**.

(115) FIGS. **20A** and **20B** illustrate IP Security headers, in accordance with some embodiments. FIG. **20A** may illustrate header **2050** that includes IP **2052**, AH (authentication) header **2054**, TCP **2056**, and data **2058**. Header **2050** may be an authentication header. FIG. **20B** may illustrate header **2060** that includes IP **2062**, encapsulating security payload (ESP) header **2064**, TCP **2066**, data **2068**, ESP trailer **2070**, and ESP authentication **2072**. Header **2060** may be an ESP header.

(116) FIGS. **21A** to **32B** illustrate various frame formats, in accordance with some embodiments. Such frame formats may be described herein, but may also be ascertained from the labels of the figures. In various embodiments, the max frame size may be, for example 192 bytes.

(117) In various embodiments, defined messages may be used as a means of communicating control plane messages as well as data plane messages between fabric manager and orchestrator, fabric manager and CXLOverEthernet Bridges, and/or fabric manager and other resources attached to a switch on chip with caches. Control plane messages may also be a way of communication among other components in the fabric of switches described herein.

(118) Various formats of packet format may be as defined as below. The packet format may communicate caching related commands through the switch fabric and between switch fabrics. Such a package format may not require the full 512 bits on internal flits (such as for PCIe) and may be treated as an additional slot format for CXL.

(119) TABLE-US-00001 Name Width Description Opcode 4 Type of operation 4'h4: Downstream Port (DSP) cache read request (SRAM destination) 4'h5: DSP cache read request (DSP destination) 4'h6: DSP cache read response to SRAM 4'h7: DSP cache read response to DSP 4'h8: DSP cache write request 4'h9-4'hF: Reserved transfer 4 Number of cache lines to be transferred size 4'h0: 1 line 4'h1: 4 lines 4'h2: 8 lines 4'h3: 16 lines 4'h4: 64 lines 4'h5-4'hF: Reserved Address 46 Memory address VH 6 Virtual hierarchy identifier- number For blocks common to different hierarchies (e.g., fabric port, accelerators) to identify the next destination fast return 1 1: indicates parallel return of the original line request to the source requester source port 12 Upstream port (SPID) Tag 16 For writes SRAM 17 1.sup.st level cache SRAM address address [16:13]: station number [12:0]: 1KB block address in 8MB station segment HDM 12 HDM decoder based decoder destination port (multiple HDM port decoder ports may share the same cache level) Wait queue 4 Waiting queue number number Prefetch 1 Indicates if prefetch request (e.g., to prevent generation of a fast return response to the upstream port)

(120) FIG. **21A** illustrates a CXL 3.0 flit for a L1 (physical) layer. Such a flit may include SKP OS

(Ordered Sets) **2102**, CXL flit #1 **2104**, SKP OS **2106**, and CXL flit #2 **2108**. As such, FIG. **21A** illustrates a CXL 3.0 flit with two different CXL flits, each preceded by SKP OS. Such flits and/or SKP OS may be different bytes of memory, as illustrated herein.

(121) FIG. **21B** illustrates a CXL 3.0 flit for a L2 layer. Such a L2 layer may be a modified L2 layer. Such a CXL 3.0 flit for a modified L2 layer may provide for increased efficiency and additional low latency traffic transport. The CXL 3.0 flit may include preamble **2122**, tag **2132** including CRC/checksum **2124**, start frame delimiter (SFD) **2126**, L2 payload **2128**, and interframe gap (IFG) **2130**. In certain embodiments, preamble **2122** may be 1 byte, tag **2132** may be 6 bytes (1 bytes payload type, 2 bytes CXL CMD/Ack/Valid/Status, 1 byte reserved, 1 byte CRC/checksum **2124**, and 1 byte flow control). The size of preamble **2122** may, in certain embodiments, be configurable. SFD **2126** may be 1 byte. L2 payload **2128** may be a payload of different bytes and may include subtags that are 1 byte. The payloads may include payload tags of the following configuration: 0: Normal Ethernet Frame 1: LL HPC (8 Bytes messages) 2: High BW HPC 3: Large Payload GPU 4: Latency sensitive GPU 5: Latency sensitive AI Traffic 6: High BW AI traffic 7: Video traffic 8: CXL2.0 Flit 9: CXL 3.0 Flit 10-12: CXL.io 13-15: CXL.\$ 16-18: CXL.mem 19: Latency sensitive AVB (Audio, Video, Broadcast) 20: Fabric manager traffic 21-255: Reserved

(122) IFG **2130** may be 1 byte. The embodiment of FIG. **21B** may additionally include 4 bits for 16 Queues PFC, 4 bits for finer level granularity. For queue occupancy, the queue may be divided into 16 parts. The size of IFG **2130** may, in certain embodiments, be configurable and, thus, may be any desired size.

(123) FIG. **22** illustrates CXL L2 frame formats for Read Request **2200**, Write Request **2220**, Write Acknowledgement **2240**, and Read Response **2260**. FIG. **22** illustrates the various components of the requests, acknowledgements, and responses, and the memory sizes thereof. The various CXL formats may include CXL CM and Control & Discovery Codes. Such codes may be defined as follows:

(124) CMD Codes:

(125) 0000_0000: Test/Sync Packet—Will be dropped by Receiver 0000_0001: Mem RdReq No Address Translation 0000_0010: Mem WrReq No Address Translation 0000_0011: Mem WrAck 0000_0100: Mem RdResp No Address Translation 0000_0101: Mem RdReq No Address Translation 0000_0110: Mem WrReq No Address Translation 0000_0111: Reserved 0000_1000: Global PM Flush 0000_1001: PM Write 0000_1010: PM Read2Sync 0000_1xxx: Reserved 0001_0000: Prefetch Read Req 0001_0001: Prefetch Read Resp 0001_0010: Prefetch Write Req 0001_0011: Prefetch Write Ack 0001_0100: Prefetch Stats Read 0001_0101: Prefetch Stats Update 0010_0001: CPU attached memory Read Req 0010_0010: CPU attached memory Read Resp 0010_0011: CPU attached memory Write Req 0010_0100: CPU attached memory Write Ack 0010_0101: CPU attached memory Stats Read 0010_0110: CPU attached memory Stats Update 0011_0001: Hot Add 0011_0010: Hot Remove 0011_0011: Device Not Responding 0011_0100: Device Uncorrectable Error 0011_0101: Device Correctable Error 0011_0110: Reserved

(126) Control & Discovery Codes: 0100_0000: Discovery 1 0100_0001: Discovery2 0100_0010: Discovery3 0100_0011: Discovery4 0100_0100: Discovery5 0100_0111: Reserved xxxx_0000: Reserved xxxx_1111: Reserved

(127) FIGS. **23A-I** illustrates various defined messages for the techniques described herein. FIGS. **24A** and **24B** may illustrate packet formats for CXL 2.5 format objects and the size thereof (with XXB=XX byte).

(128) FIG. **25** illustrates CXL 2.5 frame formats for Read Request **2500**, Write Request **2520**, Write Acknowledgement **2540**, and Read Response **2560**. FIG. **26** illustrates CXLOverMPLS frame formats for Read Request **2600**, Write Request **2620**, Write Acknowledgement **2640**, and Read Response **2660**. The frames may include up to 16 MPLS tags of 4 bytes for each tag, for a total frame size of 192 bytes. FIG. **27** illustrates CXL L3 V4 frame formats for Read Request **2700**,

Write Request **2720**, Write Acknowledgement **2740**, and Read Response **2760**. FIG. **28** illustrates CXL L3 V6 frame formats for Read Request **2800**, Write Request **2820**, Write Acknowledgement **2840**, and Read Response **2860**. FIG. **29** illustrates CXLOverMPLS V6 frame formats for Read Request **2900**, Write Request **2920**, Write Acknowledgement **2940**, and Read Response **2960**. Such a frame format may include up to 11 MPLS tags. FIG. **30** illustrates CXL L4 frame formats for Read Request **3000**, Write Request **3020**, Write Acknowledgement **3040**, and Read Response **3060**. FIG. **31** illustrates CXL MPLS L4 frame formats for Read Request **3100**, Write Request **3120**, Write Acknowledgement **3140**, and Read Response **3160**. Such a frame format may include up to 6 MPLS tags. Various, source IP may be up to 16 bytes (used as a source QN), destination IP may be up to 16 bytes (used as a destination QN), the CMD may be 1 byte, and the acknowledgement or status may be 1 byte.

(129) FIGS. **32A** and **32B** illustrate CXL 2.5 frame formats for Read Request **3200**, alternative Read Request **3210**, Write Request **3220**, Write Acknowledgement **3240**, and Read Response **3260**. The max package size for the embodiments of FIGS. **32A** and **32B** may be 320 bytes. The IPSec header may be 75 bytes for Ipv4 and 95 bytes for Ipv6. Various, source IP may be up to 16 bytes (used as a source QN), destination IP may be up to 16 bytes (used as a destination QN), the CMD may be 1 byte, and the acknowledgement or status may be 1 byte.

(130) Any of the disclosed embodiments may be embodied in various types of hardware, software, firmware, computer readable media, and combinations thereof. For example, some techniques disclosed herein may be implemented, at least in part, by non-transitory computer-readable media that include program instructions, state information, etc., for configuring a computing system to perform various services and operations described herein. Examples of program instructions include both machine code, such as produced by a compiler, and higher-level code that may be executed via an interpreter. Instructions may be embodied in any suitable language such as, for example, Java, Python, C++, C, HTML, any other markup language, JavaScript, ActiveX, VBScript, or Perl. Examples of non-transitory computer-readable media include, but are not limited to: magnetic media such as hard disks and magnetic tape; optical media such as flash memory, compact disk (CD) or digital versatile disk (DVD); magneto-optical media; and other hardware devices such as read-only memory (“ROM”) devices and random-access memory (“RAM”) devices. A non-transitory computer-readable medium may be any combination of such storage devices.

(131) In the foregoing specification, various techniques and mechanisms may have been described in singular form for clarity. However, it should be noted that some embodiments include multiple iterations of a technique or multiple instantiations of a mechanism unless otherwise noted. For example, a system uses a processor in a variety of contexts but can use multiple processors while remaining within the scope of the present disclosure unless otherwise noted. Similarly, various techniques and mechanisms may have been described as including a connection between two entities. However, a connection does not necessarily mean a direct, unimpeded connection, as a variety of other entities (e.g., bridges, controllers, gateways, etc.) may reside between the two entities.

(132) In the foregoing specification, reference was made in detail to specific embodiments including one or more of the best modes contemplated by the inventors. While various embodiments have been described herein, it should be understood that they have been presented by way of example only, and not limitation. For example, some techniques and mechanisms are described herein in the context of fulfillment. However, the disclosed techniques apply to a wide variety of circumstances. Particular embodiments may be implemented without some or all of the specific details described herein. In other instances, well known process operations have not been described in detail in order not to unnecessarily obscure the techniques disclosed herein.

Accordingly, the breadth and scope of the present application should not be limited by any of the

embodiments described herein, but should be defined only in accordance with the claims and their equivalents.

Claims

1. A system comprising: a first server device comprising: a processor; a first accelerator, separate from the processor, and configured to accelerate one or more types of workloads; a second accelerator, separate from the processor, and configured to accelerate the one or more types of workloads; a first cache coherent switch on chip, communicatively coupled to the first accelerator and the second accelerator via a Compute Express Link (CXL) protocol, wherein the first cache coherent switch on chip is configured to bypass the processor to provide cache coherency between the first accelerator and the second accelerator; and wherein the first cache coherent switch comprises one or more cache hierarchies, the one or more cache hierarchies indicating a priority for the one or more caches coupled to the first cache coherent switch.
2. The system of claim 1, further comprising: a third accelerator, wherein the first cache coherent switch on chip is further configured to provide cache coherency between first accelerator, the second accelerator, and the third accelerator.
3. The system of claim 1, further comprising: a first network interface card, communicatively coupled to the first cache coherent switch on chip and to a network.
4. The system of claim 3, wherein the first network interface card is configured to: receive, from the network, cache coherent data; and provide, to the first cache coherent switch on chip, the cache coherent data.
5. The system of claim 3, further comprising: a second server device comprising: a third accelerator; a second cache coherent switch on chip, communicatively coupled to the third accelerator and configured to: receive, from a second network interface card, the cache coherent data; and provide, to the third accelerator, the cache coherent data; and the second network interface card, communicatively coupled to the second cache coherent switch on chip and to the first network interface card via the network, wherein the first cache coherent switch on chip and the second cache coherent switch on chip are configured to provide cache coherency between the first accelerator, the second accelerator, and the third accelerator by: receiving cache coherent data from the first accelerator; providing the cache coherent data to the second accelerator; and providing the cache coherent data to the first network interface card for communication to the second cache coherent switch on chip via the network and the second network interface card.
6. The system of claim 1, further comprising a memory, wherein the first cache coherent switch on chip is further configured to provide cache coherency to the memory.
7. The system of claim 1, wherein the first cache coherent switch on chip comprises a microprocessor, and wherein the microprocessor is configured to direct cache coherent data to the first accelerator and/or the second accelerator to provide the cache coherency.
8. A method comprising: receiving, with a cache coherent switch on chip from a network interface card, cache coherent data addressed to a first accelerator, the cache coherent switch comprising one or more cache hierarchies, the one or more cache hierarchies indicating a priority for the one or more caches coupled to the first cache coherent switch; providing, by the cache coherent switch on chip to the first accelerator while bypassing the processor, the cache coherent data; receiving, with the cache coherent switch on chip from the first accelerator, a bias change; providing, by the cache coherent switch on chip to a processor, the bias change; receiving, with the cache coherent switch on chip from the processor, line resolved data; and providing, by the cache coherent switch on chip to the first accelerator, the line resolved data to cause the first accelerator to write the cache coherent data into a cache coherent memory of a second accelerator; wherein the first accelerator and the second accelerator, separate from the processor, are configured to accelerate one or more types of workloads.

9. The method of claim 8, further comprising: receiving, with the cache coherent switch on chip from the processor, snoop data; and providing, by the cache coherent switch on chip to a first snooping component, the snoop data, wherein the snoop data indicates the first snooping component.

10. The method of claim 9, wherein the first snooping component is the second accelerator.
